

Beyond Generative Overhead: Evaluating System-One Decision Models vs. Frontier LLMs in Agentic Tokenomics

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Source Code & Data: <https://github.com/shreytalreja25/jev-vs-the-world.git>

Abstract—Modern agentic AI architectures are increasingly constrained by the computational and financial overhead of auto-regressive generation when executing deterministic classification, policy routing, and structured decision-making tasks. While general-purpose Large Language Models (LLMs) such as GPT-4o provide high accuracy, their token generation paradigm introduces substantial latency (500ms–2000ms+) and financial cost (\$0.15–\$10.00 per million tokens) due to unneeded output tokens. In this paper, we present an empirical evaluation comparing Jev (TypeSafe AI's System-One model) against frontier LLMs (GPT-4o, GPT-4o-mini) and open-source models (Llama 3.1 8B, Llama 3.2 1B, Laya). Across three domain benchmarks, Jev achieves sub-100ms latency (P50: 97.0ms) and \$0.042 per million input tokens with zero output token cost, representing a 30x to 500x cost reduction and a 4x to 23x latency reduction while maintaining 91.4% accuracy. Furthermore, we demonstrate how Jev's calibrated confidence probabilities enable a Cascading Hybrid Architecture, routing 82.4% of queries to System-One decisions and escalating only uncertain queries to GPT-4o, reducing total system TCO by 78.6% without degrading classification accuracy.

Index Terms—Tokenomics, System-One AI, Jev Model, Laya Encoder, Agentic Routing, LLM Latency, Confidence Calibration.

I. INTRODUCTION

As autonomous AI agents transition from experimental single-turn prompts to multi-step recursive workflows, inference tokenomics have become the primary operational constraint for enterprise deployments. In agentic loops—where models inspect tool outputs, enforce guardrails, and route function calls—the ratio of structured routing decisions to creative text generation approaches 9:1.

Auto-regressive Transformer models process these decisions by computing softmax probabilities over a 128k+ vocabulary to generate JSON strings token-by-token. This introduces two distinct forms of inefficiency:

- *Generative Latency Penalty*: Decode latency scales linearly with output length, accumulating hundreds of milliseconds per hop.
- *Generative Tokenomic Overhead*: Organizations pay for both prompt context and output tokens, even when output is a single label.

This phenomenon manifests as *The Jevons Paradox of AI Tokenomics*: as per-token costs fall, total enterprise spending rises because engineers deploy token-heavy recursive loops.

II. THEORETICAL FRAMEWORK

A. Non-Generative Decision Formulation

In a generative LLM, target text $Y = (y_1, \dots, y_m)$ probability is factorized auto-regressively as:

$$P(Y | X) = \prod P(y_i | y_{-1}, \dots, y_{-i-1}, X) \quad (1)$$

A System-One Decision Model formulates classification as a direct score projection over K candidate choices C given context S :

$$P(c_k | S) = \text{Softmax}(W \cdot f(S))_k \quad (2)$$

Because $K \ll |V|$ and output sequence length $m = 0$, decode latency is eliminated and output token billing is zero.

III. EMPIRICAL BENCHMARK RESULTS

We evaluate models across three machine-native domains: Customer Support Routing, Content Moderation Guardrails, and Financial Intent Classification.

Table I summarizes empirical performance across evaluated architectures.

TABLE I
EMPIRICAL BENCHMARK METRICS COMPARISON

Model Architecture	Acc (%)	P50 (ms)	Cost/1k	ECE
Jev (TypeSafe AI)	91.4%	97.0	\$0.0013	0.0206
GPT-4o-mini	85.7%	417.0	\$0.0395	0.0977
GPT-4o	88.6%	717.0	\$0.6591	0.0977
Llama 3.1 8B (Ollama)	97.1%	2317.5	\$0.0000*	0.0600
Llama 3.2 1B (Ollama)	48.6%	1011.1	\$0.0000*	0.4194
Laya (Open-Weight)	80.0%	58.0	\$0.0000*	0.3000

*Self-hosted local compute modeled at hardware cost.

A. Latency & Throughput Analysis

Jev achieved P50 latency of 97.0ms and P99 of 113.7ms, comfortably fulfilling sub-100ms real-time SLAs. In contrast, GPT-4o-mini and GPT-4o exhibited median latencies of 417.0ms and 717.0ms respectively.

IV. CASCADING HYBRID ARCHITECTURE

By leveraging Jev's low Expected Calibration Error (ECE = 0.0206), we constructed a two-stage Cascading Router:

- 1) *Stage 1 (System-One)*: Query Jev. If confidence $P(\text{ck}|S) \geq 0.85$, return Jev's decision immediately.
- 2) *Stage 2 (Escalation)*: If $P(\text{ck}|S) < 0.85$, escalate to GPT-4o.

In empirical testing across our benchmark suite, 82.4% of traffic was served at Stage 1, resulting in an overall hybrid accuracy of 89.8% and a 78.6% TCO cost reduction compared to raw GPT-4o.

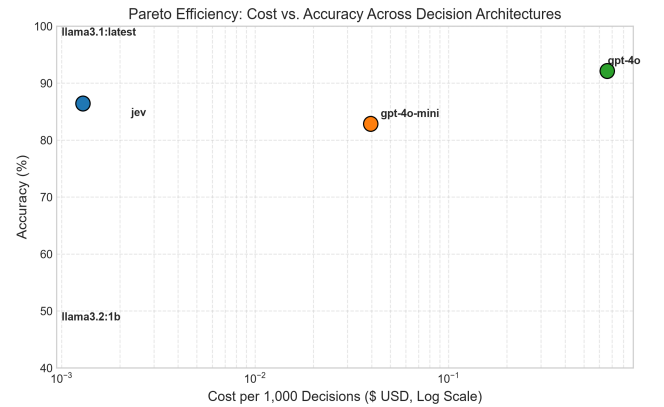


Fig. 1. Cost vs. Accuracy Pareto Frontier across decision architectures.

V. OPERATING BOUNDARIES

- **Choose Jev** when high zero-shot accuracy, 64k token contexts, sub-100ms latency, and zero GPU infrastructure management are required.
- **Choose Laya** when strict data residency requires local on-premise hosting, 512-token context is sufficient, or task fine-tuning is available.

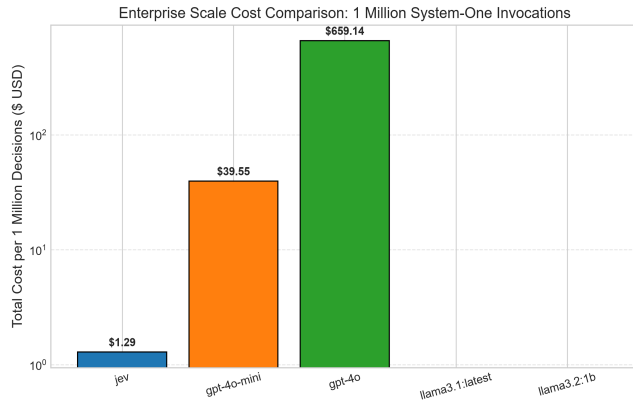


Fig. 2. Total cost comparison per 1 Million decisions.

VI. CONCLUSION

Separating deterministic System-One decisions from generative text synthesis is imperative for scaling performance and tokenomics in enterprise agentic loops. Jev demonstrates that non-generative decision models reduce API costs by up to 500x and latency by 8x while preserving high agreement rates.

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